

APPENDIX

A. Action–Context Reliance Probe

We follow the flow-time convention in the main paper, where $t = 0$ corresponds to the clean action and $t = 1$ corresponds to pure noise. For a clean action chunk a and a noise sample $\epsilon \sim \mathcal{N}(0, I)$, the flow path is

$$x_t = t\epsilon + (1 - t)a \quad (9)$$

The clean-action prediction induced by the predicted flow velocity is

$$\hat{a}_\theta(x_t, c, t) = x_t - t\hat{v}_\theta(x_t, c, t) \quad (10)$$

where the task context c includes the multi-view visual observations, the language instruction, and proprioceptive observations.

For a sampled target example i and a substitution partner j , the action-input and context intervention responses are defined as in Eqs. (3)–(4) of the main paper:

$$\delta_{\text{act}}^{(i)}(t) = \left\| \hat{a}_\theta(x_t^{(i)}, c_i, t) - \hat{a}_\theta(x_t^{(j)}, c_i, t) \right\|_2^2 \quad (11)$$

$$\delta_{\text{ctx}}^{(i)}(t) = \left\| \hat{a}_\theta(x_t^{(i)}, c_i, t) - \hat{a}_\theta(x_t^{(i)}, c_j, t) \right\|_2^2 \quad (12)$$

We first aggregate the raw replacement responses:

$$\tilde{d}_{\text{act}}(t) = \mathbb{E} \left[\delta_{\text{act}}^{(i)}(t) \right], \quad \tilde{d}_{\text{ctx}}(t) = \mathbb{E} \left[\delta_{\text{ctx}}^{(i)}(t) \right] \quad (13)$$

The response obtained when both the noised action input and task context are replaced is

$$D(t) = \mathbb{E} \left[\left\| \hat{a}_\theta(x_t^{(i)}, c_i, t) - \hat{a}_\theta(x_t^{(j)}, c_j, t) \right\|_2^2 \right] \quad (14)$$

The normalized reliance scores used in the main paper are

$$d_{\text{act}}(t) = \frac{\tilde{d}_{\text{act}}(t)}{D(t)}, \quad d_{\text{ctx}}(t) = \frac{\tilde{d}_{\text{ctx}}(t)}{D(t)} \quad (15)$$

These are normalized sensitivity measures rather than probabilities and do not generally sum to one. Since both scores are divided by the same positive quantity $D(t)$, their crossing is unchanged:

$$d_{\text{act}}(t^*) = d_{\text{ctx}}(t^*) \iff \tilde{d}_{\text{act}}(t^*) = \tilde{d}_{\text{ctx}}(t^*) \quad (16)$$

The crossover can therefore be computed equivalently from either the raw responses or the normalized scores.

For each sampled target trajectory, we draw one noise realization and keep it fixed throughout the entire flow-time sweep. The partner action input is constructed using the partner clean action and an independently sampled, fixed noise realization:

$$x_t^{(j)} = t\epsilon_j + (1 - t)a_j \quad (17)$$

Reusing ϵ_i while replacing only a_i would give

$$x_t^{(i)} - x_t^{(j, \text{shared } \epsilon)} = (1 - t)(a_i - a_j) \quad (18)$$

which forces the two action inputs to become identical as $t \rightarrow 1$ and artificially drives the action-input response toward zero.

Substitution partners are sampled from different episodes of the target task to avoid temporal correlations between nearby action chunks. A single partner assignment is drawn for each probe sweep and kept fixed across all evaluated flow times. When comparing different checkpoints, the same probe examples, partner assignments, and noise realizations are reused.

Sample-wise responses are aggregated with equal episode weighting: each episode contributes equally, and action chunks within an episode are averaged uniformly. The continuous crossover is estimated by linear interpolation between adjacent evaluated flow-time values that bracket

$$d_{\text{act}}(t) = d_{\text{ctx}}(t). \quad (19)$$

B. Task-Matched, Precision-Mismatched Boundary.

The precision-mismatch boundary t_{pm} is determined from the flow-time-dependent separability between the high-quality target-task data D_{pt} and the precision-mismatched target-task data D_{pm} . A flow-time-conditioned source classifier is trained to predict whether the noised action input originates from D_{pt} or D_{pm} . Its held-out AUC at flow time t is denoted by $\text{AUC}(t)$.

Following the criterion in the main paper, we define

$$t_{\text{pm}} = \min \{t \in [0, 1] : |\text{AUC}(t) - 0.5| \leq \sigma\} \quad (20)$$

Here, the AUC threshold is used as an operational source-separability criterion. The value $\sigma = 0.05$ is fixed across tasks and is not tuned using downstream policy performance.

To quantify the distinguishability between task-matched but precision-mismatched action distributions, we train a flow-time-conditioned binary source classifier. The classifier predicts whether a noised action sample x_t originates from D_{pm} or D_{pt} .

For each action window, we extract a relative end-effector action trajectory representation over a fixed horizon of $H = 30$ steps. The representation includes position, orientation, and gripper state, resulting in a 7-dimensional action representation consisting of *xyz*, *roll-pitch-yaw*, and gripper values at each timestep. The orientation representation is converted from quaternions to Euler angles, and angular differences are wrapped to $[-\pi, \pi]$. This representation is chosen to be consistent with the action representation used during policy training, ensuring that the classifier evaluates source differences in the same action space as the learned policy.

Given a flow time t , the clean action representation is perturbed following the flow interpolation defined in Eq. 1, producing the classifier input x_t . Before classification, the input features are standardized using statistics computed from the training set. We then train an L2-regularized logistic regression classifier with a fixed regularization coefficient ($C = 1.0$) to predict the source label. No hyperparameter tuning is performed.

The classifier is evaluated on held-out samples using the area under the receiver operating characteristic curve (AUC). By repeating the evaluation across multiple random noise realizations, we obtain the flow-time-dependent separability

Calibration step	Probe crossover t^*	Δt^*
0	0.65	–
10k	0.56	0.09
20k	0.50	0.06
30k	0.48	0.02
40k	0.46	0.02
50k	0.43	0.03
60k	0.41	0.02
70k	0.40	0.01
80k	0.39	0.01
90k	0.38	0.01
100k	0.38	0.00

TABLE VI. Evolution of the interventional action–context crossover t^* during target-only calibration on Two-Stage Cable Plugging.

curve $\text{AUC}(t)$. The precision-mismatch boundary t_{pm} is determined according to Eq. 6. Confidence intervals are estimated through episode-level bootstrap resampling, treating all windows from the same episode as a single sampling unit.

C. Precision-Matched, Task-Mismatched Boundary.

The task-mismatched-data admission boundary is derived from the relative reliance of the target policy on the noised action input and the task context. To document the behavior of this diagnostic during optimization, we evaluate the probe at intermediate checkpoints of a target-only calibration run.

The calibration policy is initialized from the pretrained $\pi_{0.5}$ checkpoint and trained using only the high-quality target-task data D_{pt} . The final checkpoint used for boundary estimation is the 50k-step checkpoint, which is also the final

checkpoint used in the downstream experiments. The 100k-step checkpoint is trained only as an additional diagnostic for examining the evolution of the probe.

For a sequence of calibration checkpoints $\{\theta_k\}_{k=0}^K$, we compute $d_{\text{act}}(t)$ and $d_{\text{ctx}}(t)$ using the same probe procedure described above. The continuous crossover at checkpoint k is defined by

$$t_k^* : d_{\text{act}}^{(k)}(t_k^*) = d_{\text{ctx}}^{(k)}(t_k^*) \quad (21)$$

At the final 50k checkpoint, the continuous crossover is

$$t^* = 0.4327 \quad (22)$$

For implementation, the diagnostic probe evaluates flow time on a grid with spacing $\Delta t = 0.1$. We map the continuous crossover to the largest grid value not exceeding it:

$$t_{\text{tm}} = \Delta t \left\lfloor \frac{t^*}{\Delta t} \right\rfloor = 0.4 \quad (23)$$

The operational admission region for task-mismatched data is therefore

$$T_{\text{tm}} = [0, 0.4] \quad (24)$$

The diagnostic grid is used only for estimating and discretizing the boundary. During policy training, the flow time is sampled uniformly from the corresponding admission region, as described in Sec. III-D.

Table VI reports the evolution of the interventional crossover during target-only calibration on Two-Stage Cable Plugging. Values are reported to two decimal places in the table; the unrounded value at the final 50k checkpoint is 0.4327.